

CSCI E-102 Econometrics and Causal Inference with R
Part I of Assignment 7

Sally needs to know whether her average hourly earnings will be different next year (in current U.S. dollars). She decides to run the following econometric model:

$$\text{ahe}_i = \beta_0 + \beta_1 \text{yrseeduc}_i + \beta_2 \text{age}_i + u_i,$$

where

ahe_i = Average Hourly Earnings in current U.S. dollars;

yrseeduc_i = Years of Education;

age_i = Age (in Years);

and gets the output:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept) -18.443636   0.330477 -55.809 < 2.2e-16 ***
yrseeduc      2.340936   0.021318 109.811 < 2.2e-16 ***
age           0.189979   0.004124  46.066 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

1. Assuming these estimates of the causal effects can be considered unbiased and consistent, how much more is she expected to make (in current U.S. dollars) next year, given she studies?

Hint: You need to find $\Delta \widehat{\text{ahe}}_i$ due to simultaneous $\Delta \text{yrseeduc}_i = 1$ and $\Delta \text{age}_i = 1$.

2. Find the standard error $\text{SE}(\Delta \widehat{\text{ahe}}_i)$ of the estimated change if the variance matrix $\Sigma_{\hat{\beta}}$ of $\hat{\beta} = (\hat{\beta}_0 \hat{\beta}_1 \hat{\beta}_2)^\top$ is estimated to be

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> vcovHC(res, type = "HC0")
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              (Intercept)      yrseeduc      age
(Intercept)  0.1092152129 -6.053495e-03 -7.189159e-04
yrseeduc     -0.0060534950  4.544468e-04  3.384433e-06
age          -0.0007189159  3.384433e-06  1.700778e-05

```

$$1) \Delta \hat{a}_{he_i} = \hat{\beta}_{1i} (\Delta y_{rseduc_i}) + \hat{\beta}_{2i} (\Delta age_i)$$

$$\Delta \hat{a}_{he_i} = 2.340936(1) + 0.189979(1)$$

$$\Delta \hat{a}_{he_i} = 2.530915 \leadsto \boxed{+ \$2.53/\text{hour}}$$
 is Sally's
 Δa_{he} given that she studies
 one more year

$$2) SE(\Delta \hat{a}_{he_i}) = \sqrt{d^T \underbrace{\hat{\Sigma}_{\hat{\beta}}}_{\text{covariance matrix}} d} \quad \begin{aligned} d^T &= (0 \ 1 \ 1) \\ d &= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \end{aligned}$$

$$SE(\Delta \hat{a}_{he_i}) = \sqrt{(0 \ 1 \ 1) \begin{bmatrix} 0.1092152129 & -0.006053495 & -0.0007189159 \\ -0.006053495 & -0.0004544468 & -0.00003384433 \\ -0.0007189159 & -0.00003384433 & -0.0001700778 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}$$

$$SE(\Delta \hat{a}_{he_i}) = \sqrt{(-0.0004544468 + -0.00003384433 + -0.00003384433 + -0.0001700778)}$$

$$SE(\Delta \hat{a}_{he_i}) = \sqrt{0.000478223446}$$

$$SE(\Delta \hat{a}_{he_i}) = 0.0218683206$$